Jiefu (Jeff) Zhang

zhan4018@purdue.edu | jefferyz001.github.io | github.com/jefferyz001 | West Lafayette, IN

Research Interests

AI for chip design and manufacturing automation: reusable, learning-based physical design (macro placement); reliable evaluation of ML-for-EDA results; LLM- and agent-driven design flows; hands-on RTL-to-layout VLSI/ASIC design.

Education

Purdue University West Lafayette, IN
Master of Science in Computer Engineering (Thesis Track) *Expected December 2026*
Bachelor of Science in Computer Engineering *Graduated December 2024*

- Accelerated BS/MS Program | Graduate GPA: 3.88/4.0
- **Relevant Coursework:** ASIC Design, VLSI Design, Semiconductor Fabrication, RISC-V Architecture, Embedded Systems, Artificial Intelligence

Honors & Awards

- **NeurIPS 2026 Financial Assistance Award** – full conference and workshop registration and hotel stay

Publications & Preprints

- **J. Zhang**, H. Sun, Y. Xu, V. Aggarwal, Z. Wan, "CARVE: Breaking Data Barriers in Chip Placement by Harnessing Reusable Expertise." *arXiv:2609.33106*, 2026.
- **J. Zhang**, Y. Xu, H. Sun, Z. Zhou, T. Cao, V. Aggarwal, "Towards Reliable LLM Evaluation: Correcting the Winner's Curse in Adaptive Benchmarking." ***NeurIPS 2026***. *arXiv:2605.05973*.
- H. Sun, **J. Zhang**, Y. He, Y. Xu, V. Aggarwal, B. Bhargava, A. L. Liu, "Reliable Replay through Spatial Coherence in Online Continual Learning." *arXiv:2609.33725*, 2026.
- Y. Xu, C. Li, **J. Zhang**, H. Sun, Z. Li, V. Aggarwal, "Selection-Aware Stress Testing for Interactive Agents." *arXiv:2608.30916*, 2026.
- **J. Zhang**, Y. Xu, V. Aggarwal, "Don't Freeze, Don't Crash: Extending the Safe Operating Range of Neural Navigation in Dense Crowds." *arXiv:2603.06729*, 2026.
- H. Sun, Y. Xu, **J. Zhang**, X. Wu, Z. Zhou, J. He, J. Chen, "Core-Halo Decomposition: Decentralizing Large-Scale Fixed-Point Problems." *arXiv:2605.08681*, 2026.

Research Experience

Graduate Research Assistant, CLAN Lab, Purdue University Oct 2024 – Present
Advisor: Prof. Vaneet Aggarwal

MS Student Member, Wan Lab, Columbia University Jun 2026 – Present
Advisor: Prof. Zishen Wan

CARVE: Reusable Expertise for Chip Macro Placement (first author) arXiv:2609.33106

- Developed a framework that adapts pretrained RL macro-placement policies to new circuits when the provider's original training data are unavailable. A frozen base policy and saved specialists are reused only on tasks where they pass local validation; a new specialist is trained from the frozen base only when none qualifies.
- The framework comes with expected-performance guarantees for repeated reuse and matching bounds on the number of local samples needed for reliable reuse.
- On IBM placement benchmarks, a reuse-first variant cut recorded training time by 58.5% (9.66 $\rightarrow$ 4.01 h) while mean HPWL gain moved only from 8.41% to 7.86%; a simulated receiving operator reused imported specialists on 6 of 7 new IBM circuits without training of its own.
- Joint project of CLAN Lab (Purdue) and Wan Lab (Columbia University).

SIREN: Correcting the Winner's Curse in LLM Evaluation (first author) **NeurIPS 2026**

- Co-developed a repeated-split reporting protocol that corrects the winner's curse when a tuned configuration is selected and reported on the same benchmark data: it freezes a post-search shortlist, separates selec-

tion from held-out evaluation, and gives simultaneous confidence bands through an item-level multiplier bootstrap.
- Contributed to the theory showing that the repeated-split estimator admits a first-order item-level representation, which justifies bootstrap validity on a finite budget grid.
- Led MMLU-Pro tuning experiments on open-weight models (vLLM serving, DSPy prompt optimization), showing that winner-based reporting is optimistic and can flip deployment conclusions while SIREN stays close to the finite-sample target.
- Ongoing: extending SIREN to AI-for-EDA results, where the best run of a learned optimizer is typically selected and reported on the same benchmark circuits; current experiments use macro placement.

SPHERE: Reliable Replay through Spatial Coherence in Online Continual Learning arXiv:2609.33725
- Co-developed a replay-allocation method that aggregates signed prospective loss changes through a representation kernel, damping isolated spikes while keeping coherent increases, then allocates replay by entropy-regularized transport blended with uniform rehearsal. Evaluated on noisy-label vision tasks, continual instruction tuning of language models, and code-generation RL with incomplete test rewards.

Safe Robot Navigation in Dense Crowds (PSS-Social) (first author) arXiv:2603.06729 · code
- Built a PPO-based navigation framework with density-invariant observations, density-randomized training and density-adaptive proxemic reward shaping to address the freezing-robot problem; reached 86%+ safe success at 21-agent crowd densities where DS-RNN and SARL baselines collapse, with ORCA and SFM as analytical baselines.

Core-Halo Decomposition for Decentralized Fixed-Point Problems arXiv:2605.08681
- Built SUMO and PettingZoo traffic-control benchmarks on a 14-intersection bridge and a 7-intersection ring, plus the decentralized DQN pipeline, showing Core-Halo closes most of the gap to centralized DQN in mean waiting time with only one-hop read access; contributed the PageRank and smart-grid experiments.

Selected Hardware Projects

Multicore RISC-V Processor
- Designed a 5-stage pipelined RISC-V processor with hazard resolution through forwarding and stalling; improved multicore efficiency by 30% with the MOESI cache-coherence protocol.

USB 1.1 ASIC Data-Transfer Engine
- Implemented a 12 Mbps full-speed USB 1.1 data-transfer engine in Verilog with an AHB-Lite interface; verified in QuestaSim with 98% test coverage.

8-bit Wallace Tree Multiplier ECE 559 MOS VLSI Design
- Full-custom schematic and layout in Cadence Virtuoso, DRC/LVS clean, with parasitic extraction for timing and power; worst-case delay $\leq$ 2.5 ns, energy $\leq$ 850 fJ.

Teaching Experience

Graduate Teaching Assistant, ECE 33700 ASIC Design Laboratory, Purdue Jan – May 2025
- Instructed 60+ students in ASIC design; authored 10+ Verilog-based homework and lab assignments.

Technical Skills

- **Programming:** Python, C/C++, Verilog, SystemVerilog, MATLAB, Java
- **ML / AI for EDA:** PyTorch, deep RL (PPO, DQN), RL-based macro placement, continual learning, LLM and agent evaluation, DSPy, vLLM, Gym, PettingZoo, SUMO
- **Hardware / EDA:** Cadence Virtuoso, QuestaSim, DRC and LVS, parasitic extraction, RTL design, FPGA, ASIC and SoC integration
- **General:** Linux, Git, LaTeX

Additional

- **Industry:** Product Maintenance Engineer Intern, Compound Co., Shenzhen (Summer 2023) – sensor calibration and IoT integration for medical devices.
- **Languages:** English (fluent), Mandarin (native)